Hallucination Failure Types

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

Week 6: a citation must support its attached claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Explain the mechanism and terminology of generation failures and verification. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of generation failures and verification.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 14: Hallucination Failure Types

Fluent **output** may contain **unsupported claim**s, fabricated detail, or **faulty inference**.



CREATOR MISSION

Tag claims in an answer and classify each failure type.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Recall Controlled prompt experiments. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

This week's end goal

By the end, I can distinguish supported, **contradicted** and **unsupported** generated claims.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Hallucination Failure Types case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use hallucination failure types to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Generated answers may invent facts, misread quantities or cite irrelevant passages. Verification checks each claim against **evidence** rather than accepting fluent wording.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

STUDENT LESSON · 3 OF 18

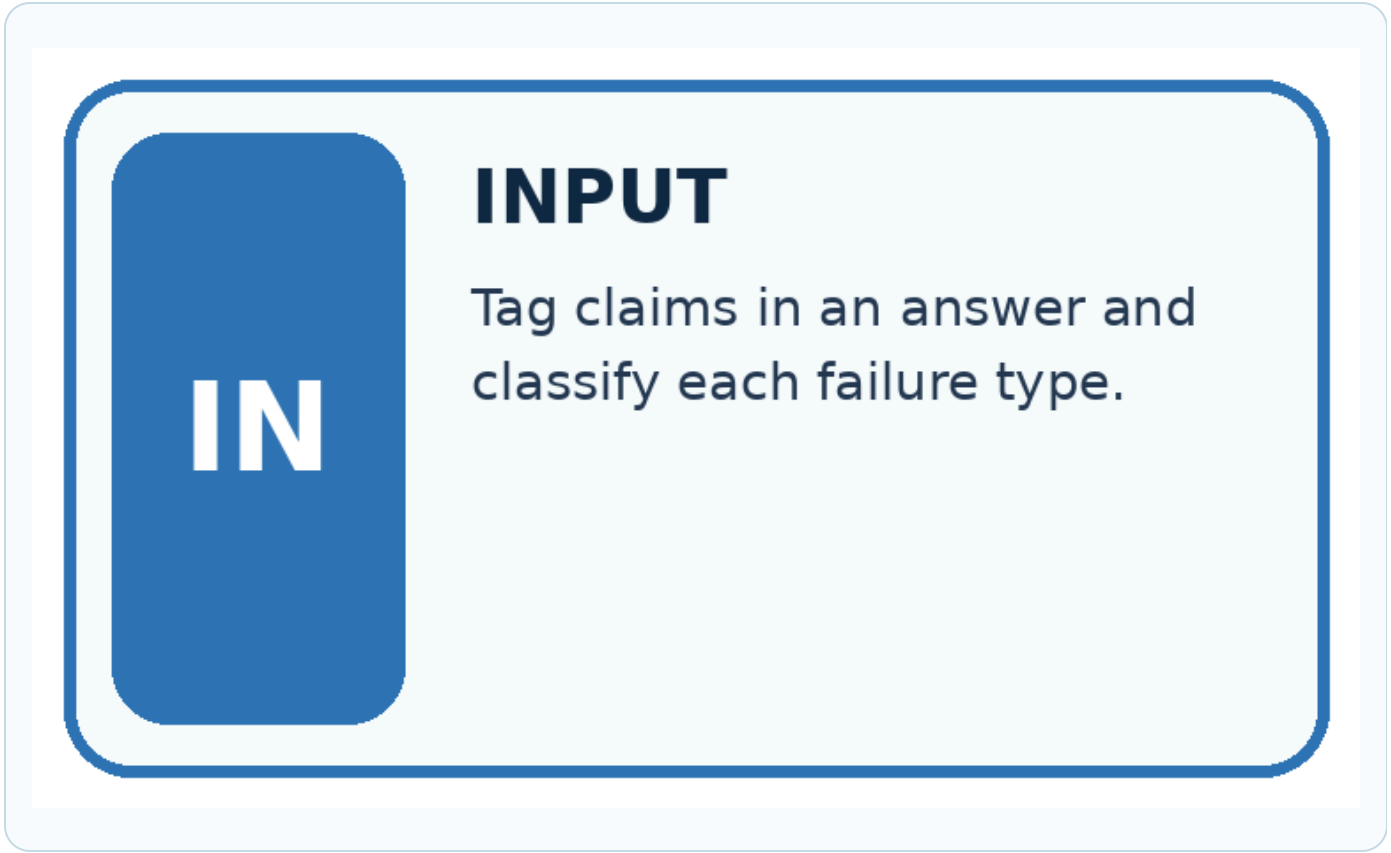
See how it works

> Follow the mechanism for Hallucination Failure Types. Connect each labelled input, deciding step and result.

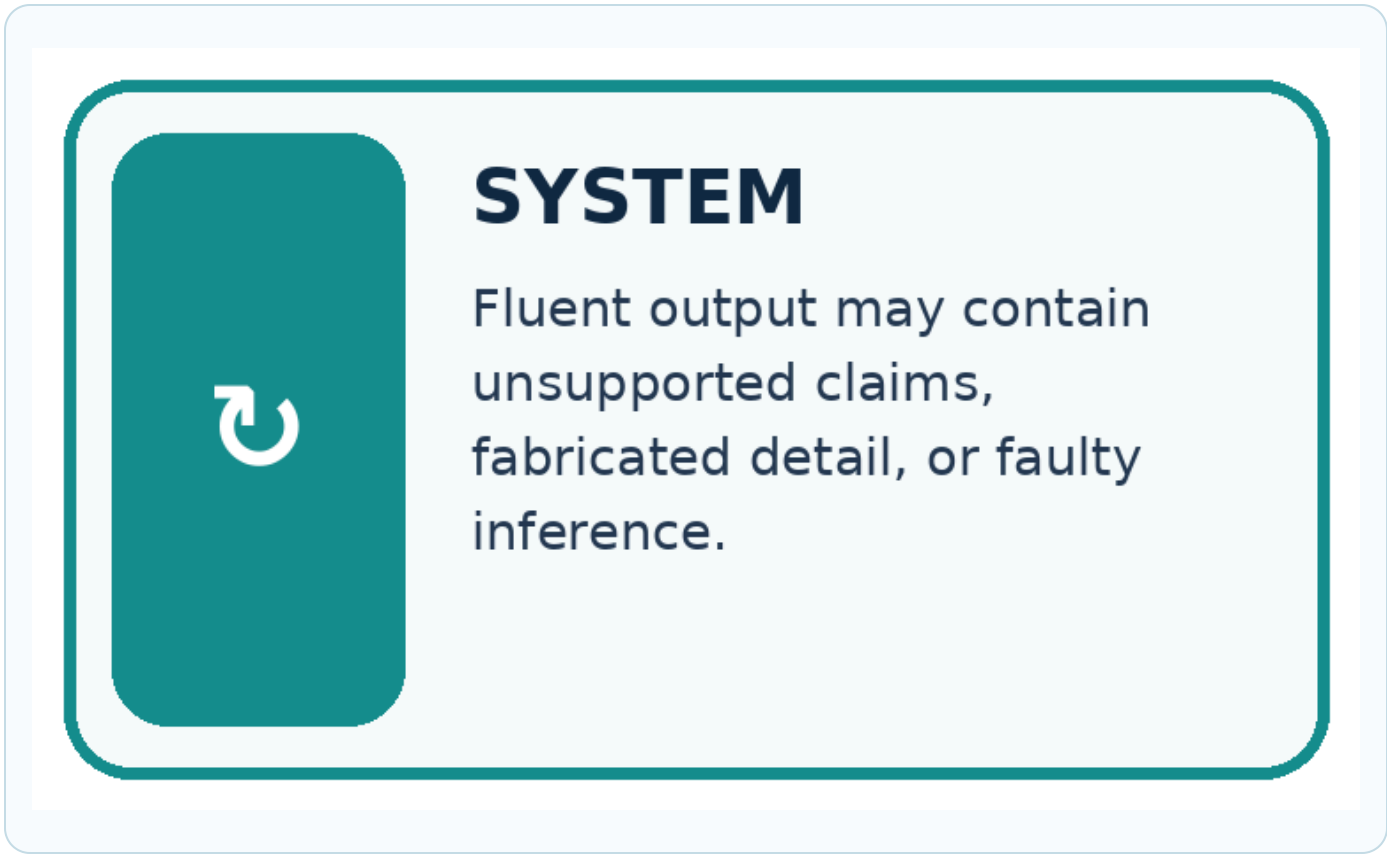
VISUAL MECHANISM

Input → Process → Output

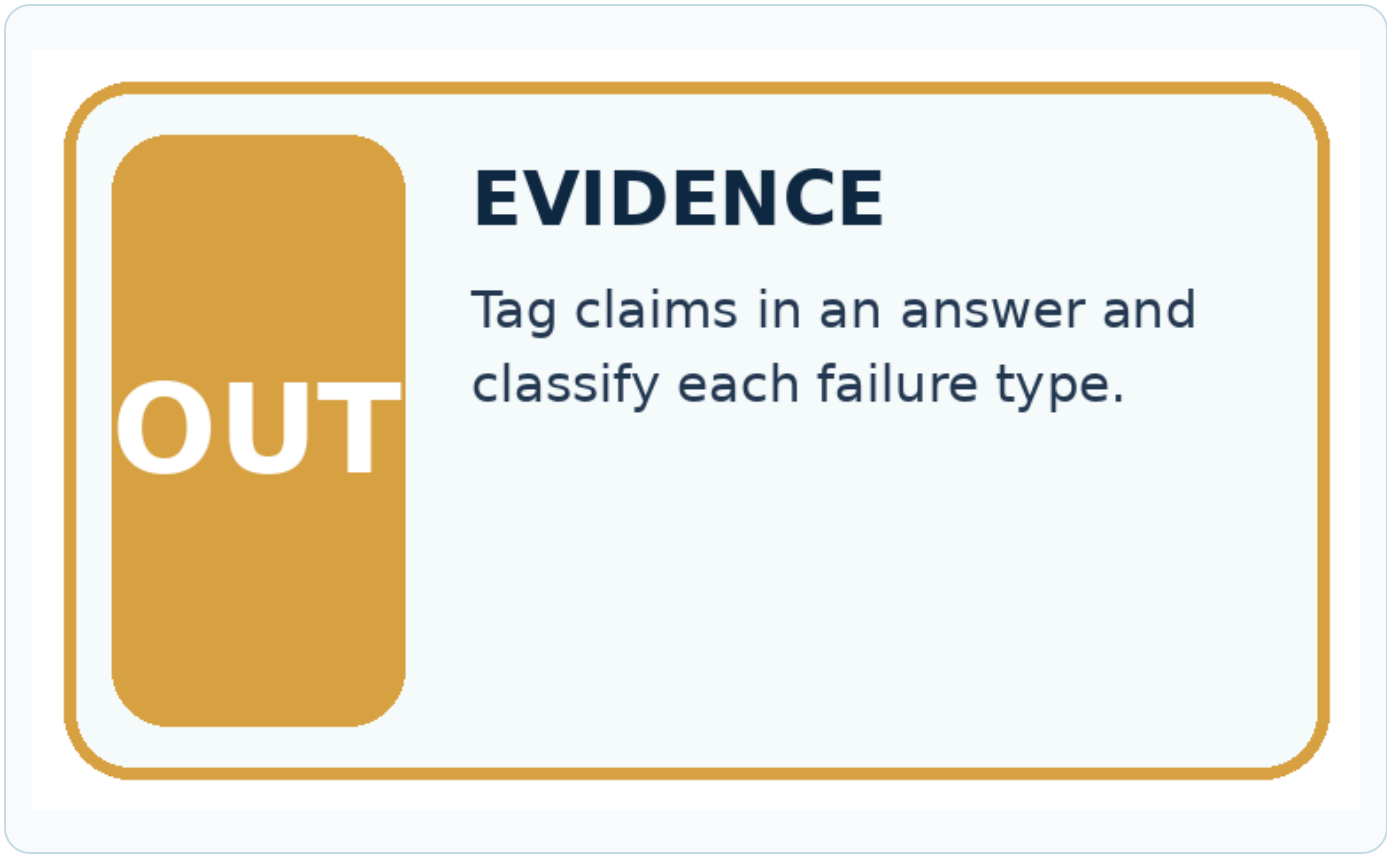
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Fluent output may contain unsupported claims, fabricated detail, or faulty inference.

↓


OUT
EVIDENCE
Tag claims in an answer and classify each failure type.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Split an answer into **atomic claim**s. For each, record the exact source passage, support status and any needed calculation. Mark **unsupported** and **contradicted** claims separately; do not repair them by inventing sources.

Generation failures and verification — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.	Split an answer into atomic claims. For each, record the exact source passage, support status and any needed calculation. Mark unsupported and contradicted claims separately; do not repair them by inventing sources.	Capacity 20 is contradicted; 15:00 is supported; free lunch is unsupported. One of three claims has support. The missing lunch fact is not evidence of no lunch.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

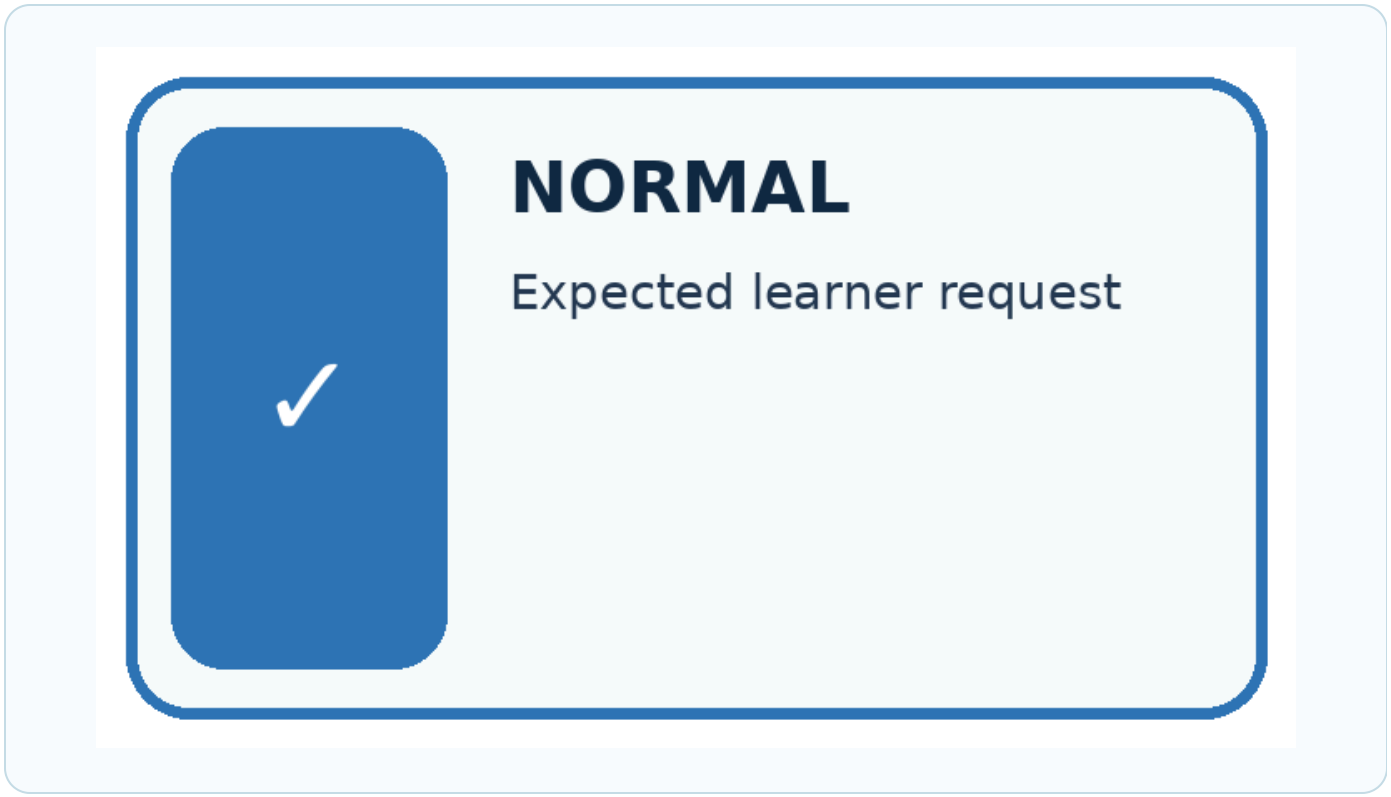
> Compare the supplied cases for Hallucination Failure Types. Identify the detail that changes the conclusion.

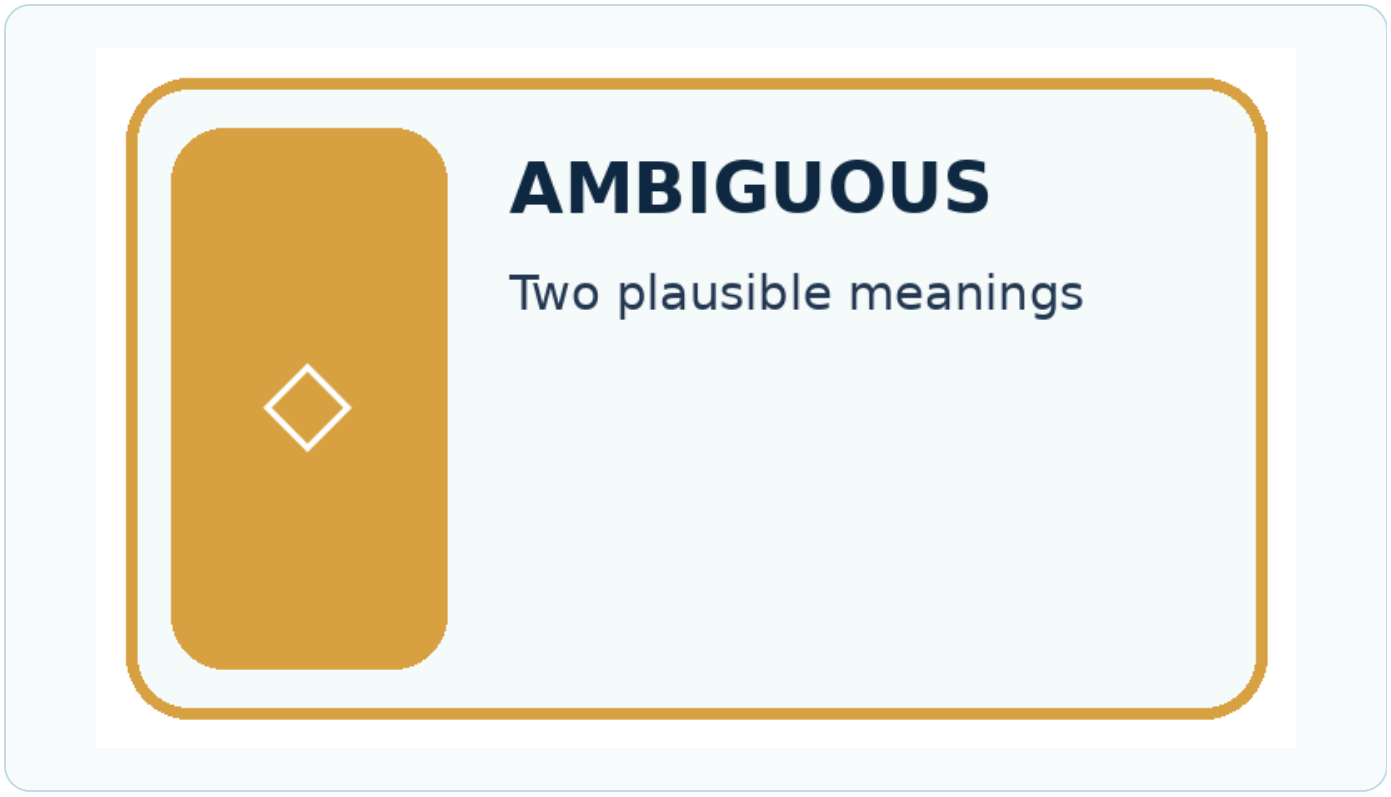
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Add a planned 30-minute stop while preserving speed and distance.

Predicted result

Prediction: Planned total elapsed time becomes 3.5 hours.

Why the result follows

Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Hallucination Failure Types to explain a concrete decision, rather than only memorising definitions.

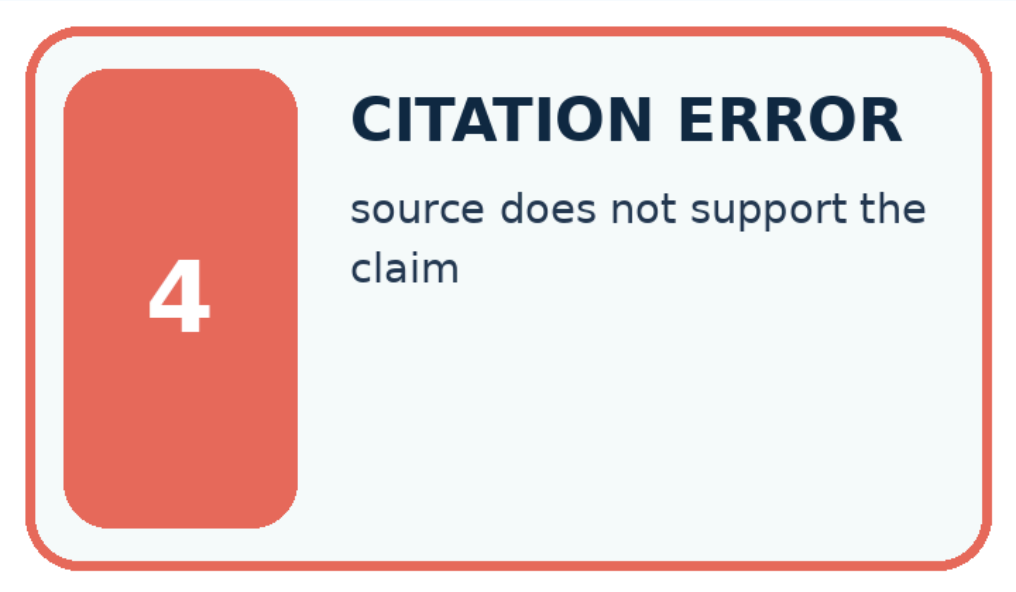
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
UNSUPPORTED CLAI
statement lacking evidence
2
FABRICATION
invented detail or source





Word	My own example	Not this / boundary
unsupported claim	A statement for which the supplied evidence does not establish the asserted conclusion.	
fabrication	Invented information presented as if it were established fact.	

Word	My own example	Not this / boundary
faulty inference	A conclusion that does not follow from the available premises or evidence.	
citation error	A reference that is wrong, missing or does not support the claim attached to it.	

Words used in the closer explanation

Word	Meaning
Atomic claim	A single factual statement checked separately
Contradicted	Inconsistent with the supplied evidence
Unsupported	Not established by the available evidence

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Hallucination Failure Types, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 14 • WORKED EVIDENCE

1. Claim 1 is directly supported by the passage.
2. Claim 2 invents a scientist’s quotation: fabrication.
3. Claim 3 draws a cause from correlation: faulty inference.
4. Tag each claim, repair only with evidence, or mark unknown.

1. Claim 1 is directly supported by the passage.

2. Claim 2 invents a scientist’s quotation: **fabrication**.

3. Claim 3 draws a cause from correlation: **faulty inference**.

4. Tag each claim, repair only with **evidence**, or mark unknown.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Source R says room B seats 12 and the meeting starts at 15:00. Generated answer says B seats 20, starts at 15:00 and includes free lunch.

Capacity 20 is **contradicted**; 15:00 is supported; free lunch is **unsupported**. One of three claims has support. The missing lunch fact is not **evidence** of no lunch.

Worked explanation · Hallucination Failure Types

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Hallucination Failure Types

A fictional Study Buddy drafts a field-trip summary from one approved record.

The facts we will use

Supplied facts

Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Supplied facts

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

AI ACADEMY / AI-3/14

Hallucination Failure Types

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute duration

12 / 4=3 hours, excluding stops.



2. Contradicted claim

Two-hour duration disagrees with the calculation from S.



3. Supported claim

Departure 09:00 is stated in S.



4. Unsupported claim

Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

CHANGE ONE THING

Add a planned 30-minute stop while preserving speed and distance.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute duration	12÷4=3 hours, excluding stops.
2. Contradicted claim	Two-hour duration disagrees with the calculation from S.
3. Supported claim	Departure 09:00 is stated in S.
4. Unsupported claim	Free lunch has no source; missing evidence is not a confirmed no-lunch fact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

What this example does not establish

Keep the stated limit

The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

Keep the stated limit

Read the labelled picture: Can we conclude there will definitely be no lunch from S alone?

Keep the stated limit

Change one thing: Add a planned 30-minute stop while preserving speed and distance. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Hallucination Failure Types claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest **unsupported** assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A confident, detailed answer is less likely to hallucinate. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

STUDENT LESSON · 8 OF 18

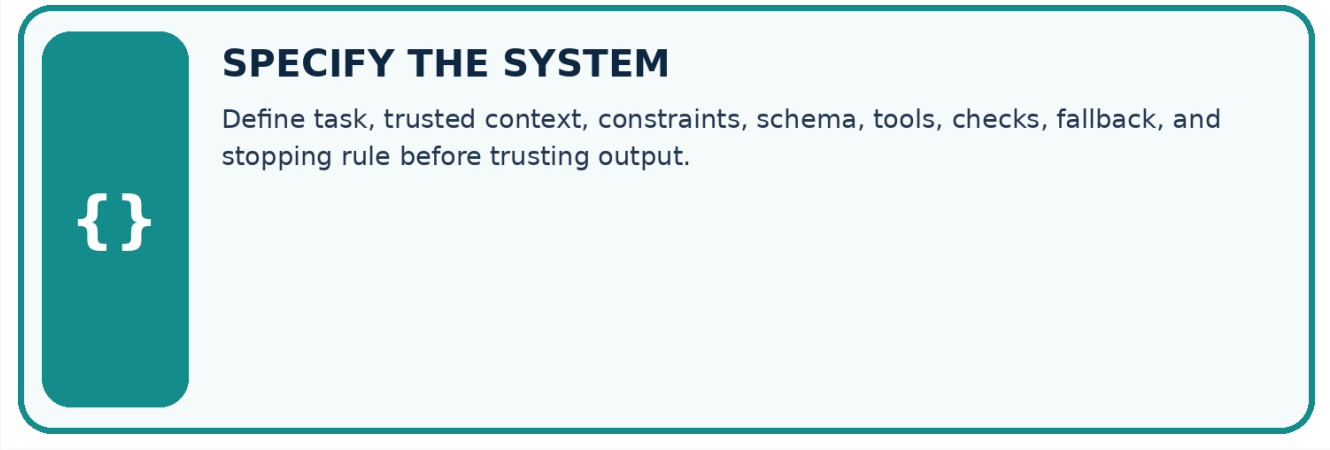
Practise together

> Practise Hallucination Failure Types with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Source agreement and arithmetic validity are separate checks; missing **evidence** is not proof of the opposite claim.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Generation failures and verification”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Generation failures and verification. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Can we conclude there will definitely be no lunch from S alone?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

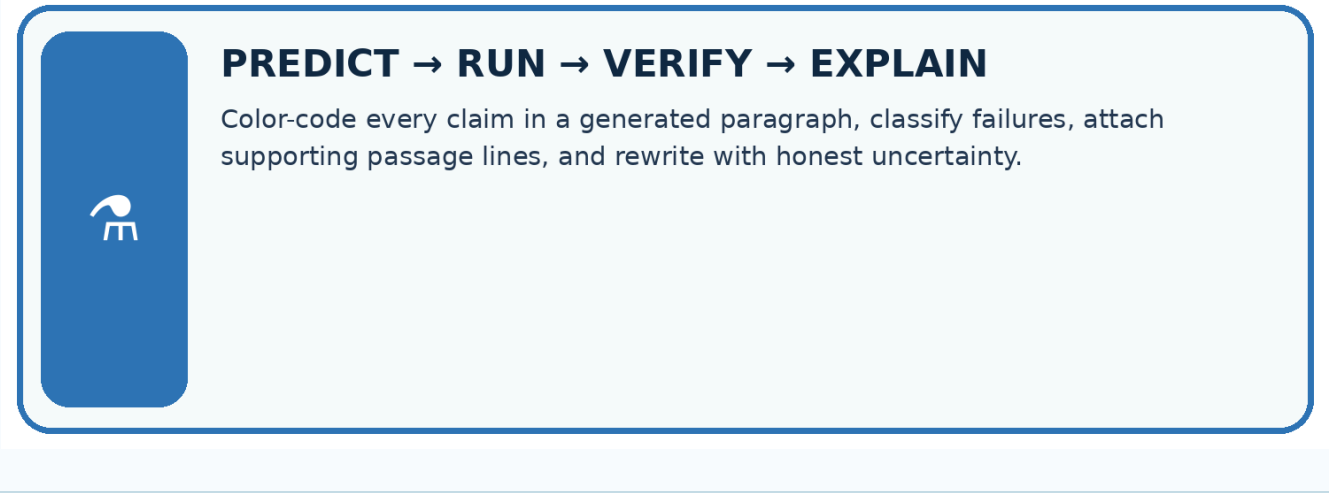
Run the investigation

> Investigate Hallucination Failure Types. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Color-code every claim in a generated paragraph, classify failures, attach supporting passage lines, and rewrite with honest uncertainty.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

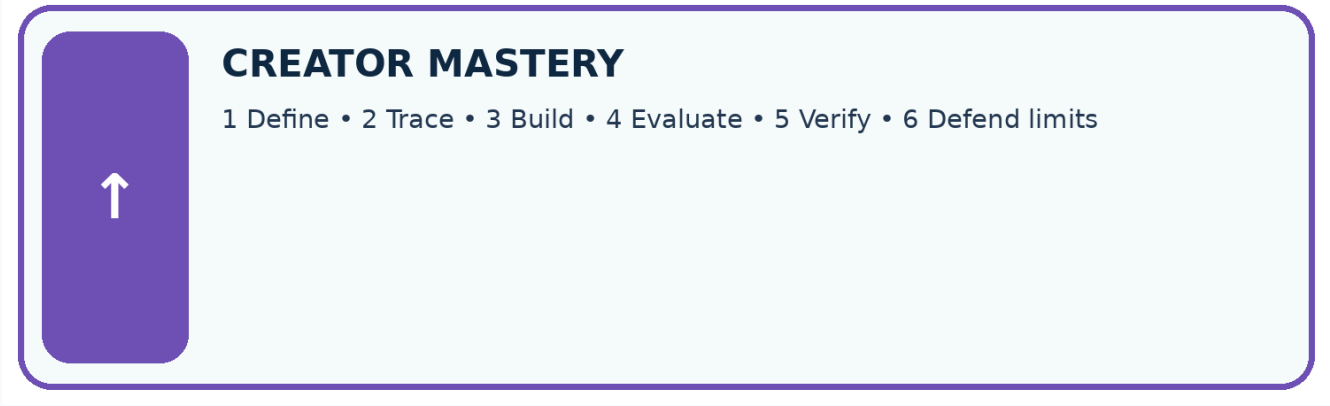
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Hallucination Failure Types. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

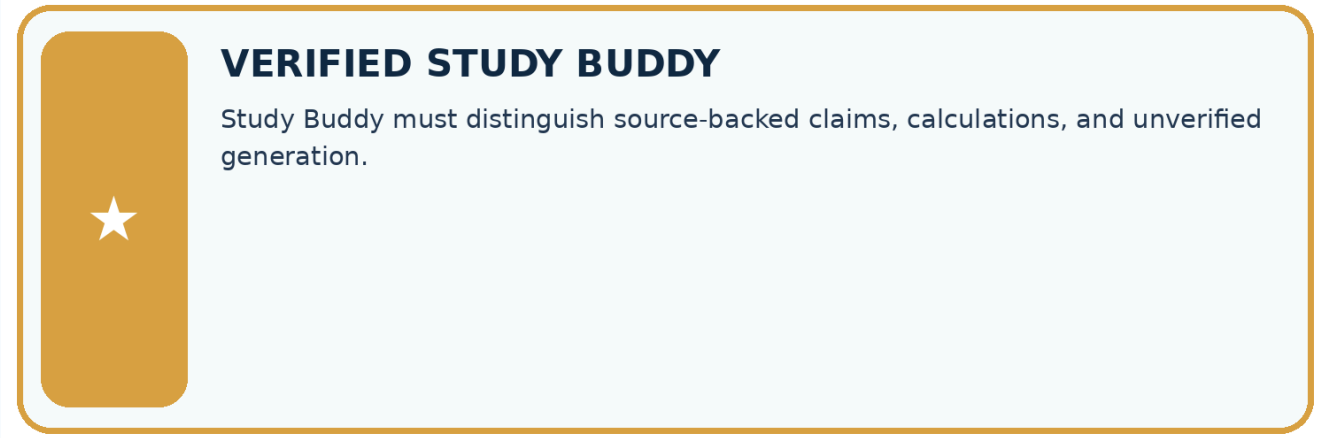
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately.

STUDENT LESSON · 12 OF 18

Show what you know

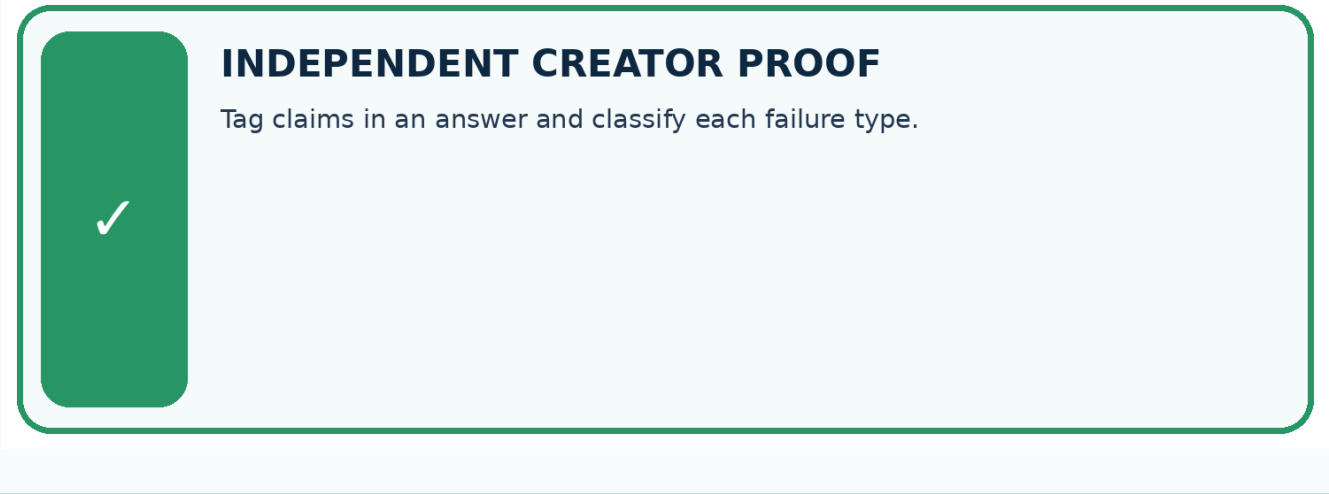
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Tag claims in an answer and classify each failure type.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Hallucination Failure Types case for supported practice. Repeated examples do not become fresh assessments.

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

The question

Separate the failures and checks.

Worked explanation

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Different failures require different checks

Evidence or stage	Value or result
Quotation	Locate exact source text
Citation	Check support for the claim
Date	Check source and calendar consistency
Repair	Correct or remove unsupported statements

An answer invents a quotation, cites a source lacking it and supplies an impossible date.

Separate the failures and checks.

Check exact quote text, claim-to-source support and calendar consistency independently. Each failure needs its own correction.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Source S: trip is 18 km; vehicle speed is 6 km/h, constant, excluding stops. Answer says 2 hours, 18 km and includes lunch. Audit and recalculate.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An answer gives a book title that does not exist and quotes words absent from the cited passage. Identify two failures and a repair.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Study Buddy must distinguish source-backed claims, calculations, and unverified generation.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A generated answer contains a nonexistent quote, a real source unrelated to the claim and a correct sum using the wrong **input**. Classify the three failures and their repairs.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first **unsupported** step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Hallucination Failure Types, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional Study Buddy drafts a field-trip summary from one approved record.

By the end, I can	How I will show it
Explain the mechanism and terminology of generation failures and verification	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Hallucination Failure Types without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of generation failures and verification

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy drafts a field-trip summary from one approved record.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.

1. **Unsupported claim**

2. Supported claim

3. **Contradicted** claim

4. Compute duration

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: **Unsupported claim** Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact. Different errors need different repairs.

Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add a planned 30-minute stop while preserving speed and distance. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Planned total elapsed time becomes 3.5 hours.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Planned total elapsed time becomes 3.5 hours. Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Repair a missing building block

Review Multimodal **Evidence** (ai-3/13). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Why were both “exactly three” and “exactly five” **unsupported** before opening the box?. Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Explain the **mechanism** and terminology of image generation and provenance. Short practice: Which checks pass after that single change, and what still needs repair?.

Return to this week and explain what you changed.

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

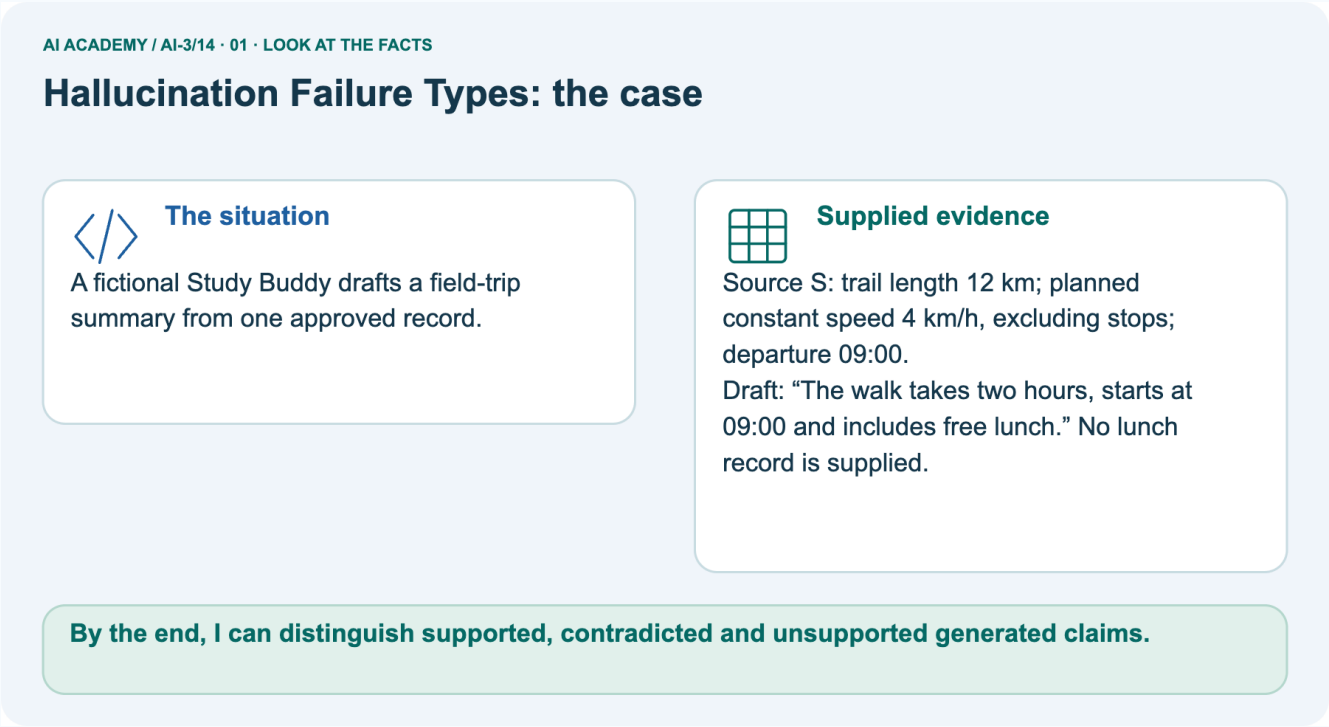
Drafts stay in this browser when storage is available; clear drafts on a shared device.

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Visual concept journey

> Explain each Hallucination Failure Types graphic using its labels, deciding evidence and relevant limitation.

01. Hallucination Failure Types: the case



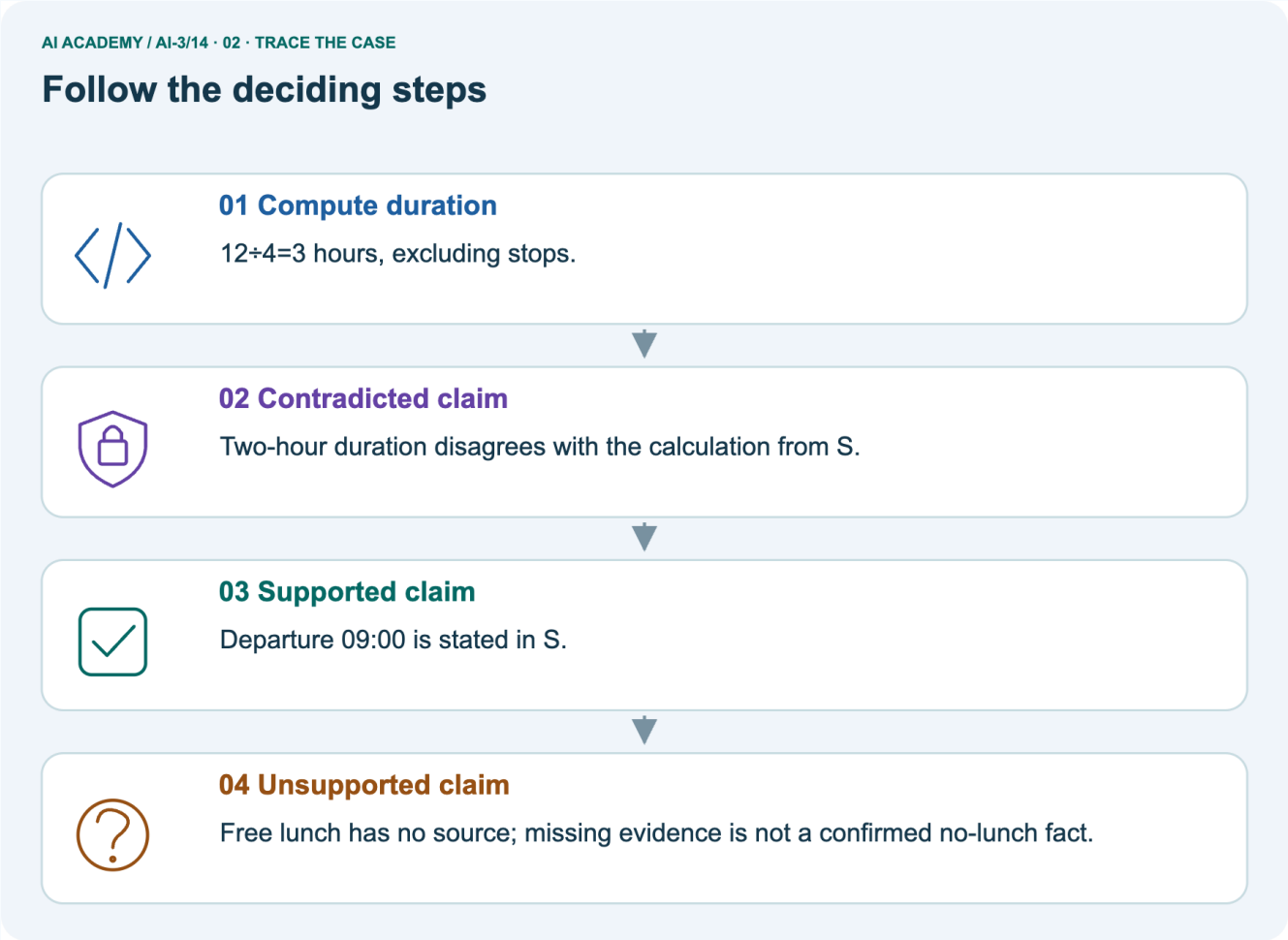
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy drafts a field-trip summary from one approved record. Supplied facts: Source S: trail length 12 km; planned constant speed 4 km/h, excluding stops; departure 09:00.

Draft: “The walk takes two hours, starts at 09:00 and includes free lunch.” No lunch record is supplied.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

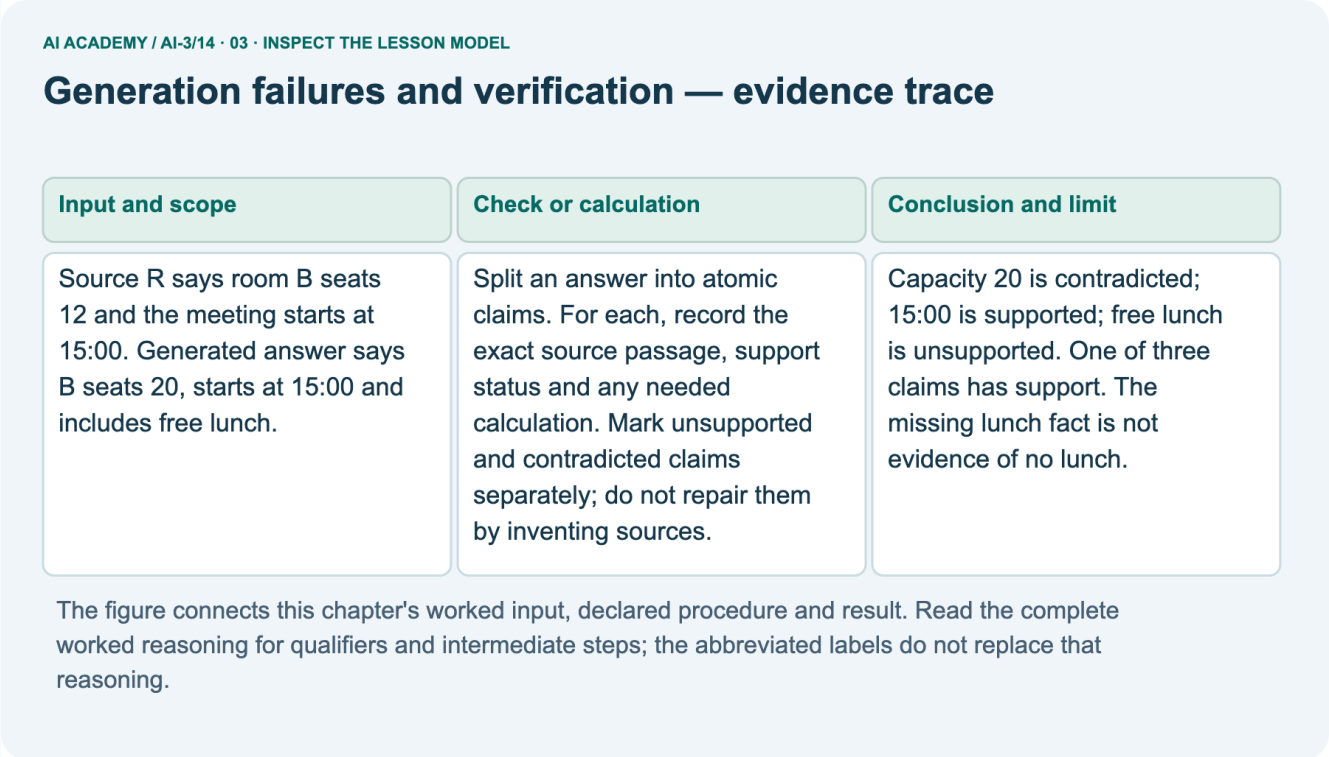


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute duration: 12÷4=3 hours, excluding stops. → **Contradicted** claim: Two-hour duration disagrees with the calculation from S. → Supported claim: Departure 09:00 is stated in S. → **Unsupported claim**: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Generation failures and verification — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generation failures and verification — **evidence** trace. Source R says room B seats 12 and the meeting starts at 15:00.

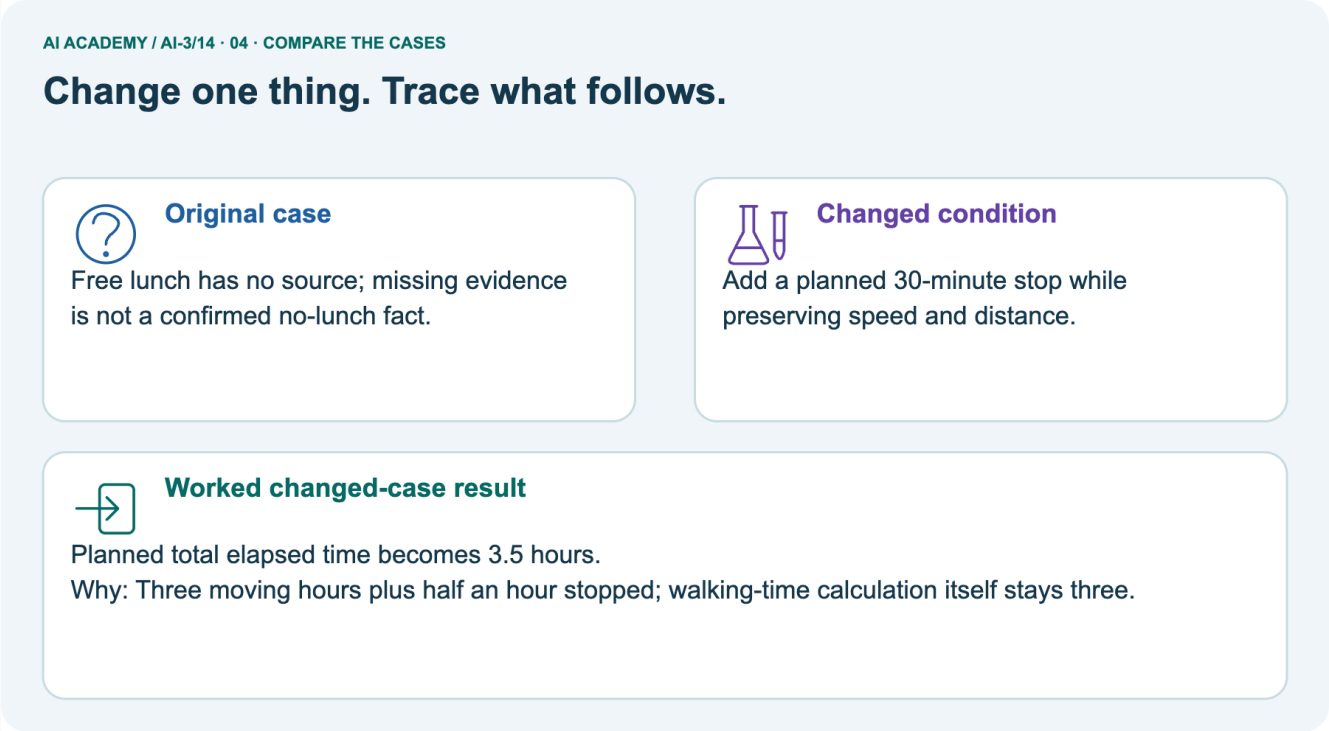
Generated answer says B seats 20, starts at 15:00 and includes free lunch. | Split an answer into **atomic claim**s. For each, record the exact source passage, support status and any needed calculation.

Mark **unsupported** and **contradicted** claims separately; do not repair them by inventing sources. | Capacity 20 is **contradicted**; 15:00 is supported; free lunch is **unsupported**. One of three claims has support.

The missing lunch fact is not **evidence** of no lunch.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



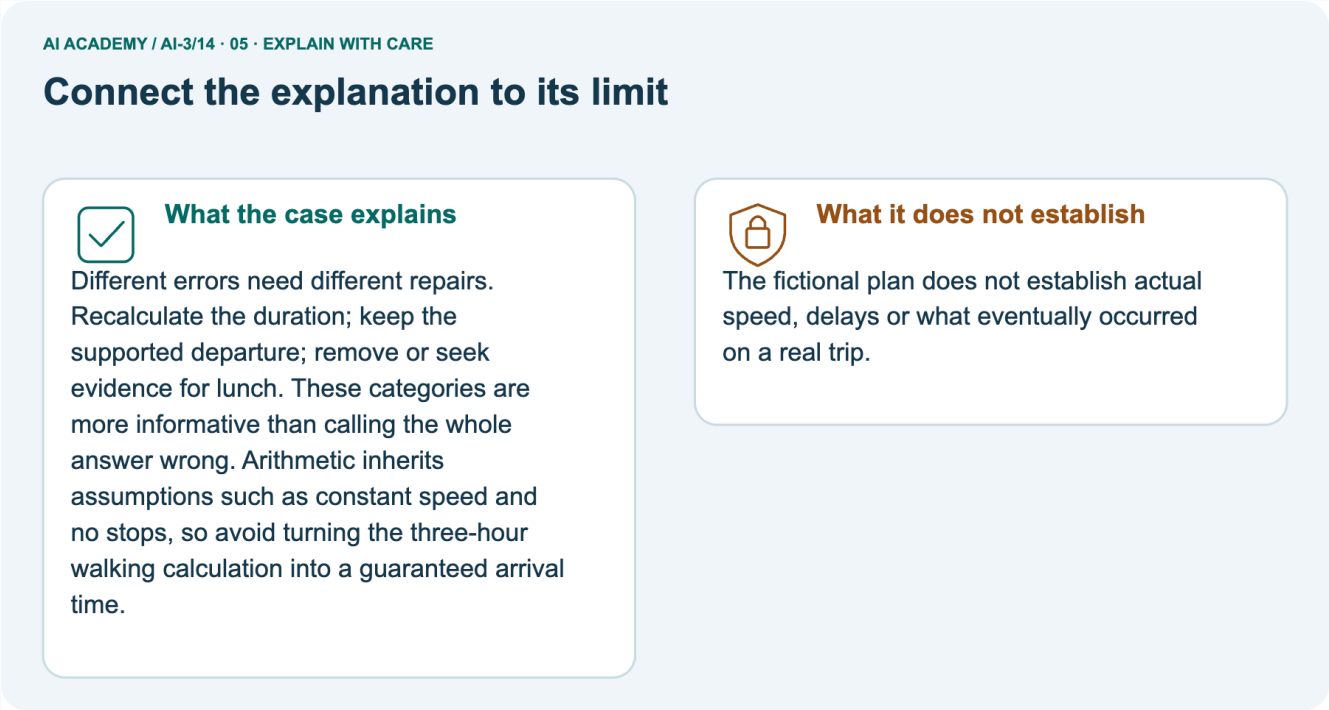
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Free lunch has no source; missing **evidence** is not a confirmed no-lunch fact.
Changed condition: Add a planned 30-minute stop while preserving speed and distance.

Expected result: Planned total elapsed time becomes 3.5 hours. Reason: Three moving hours plus half an hour stopped; walking-time calculation itself stays three.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

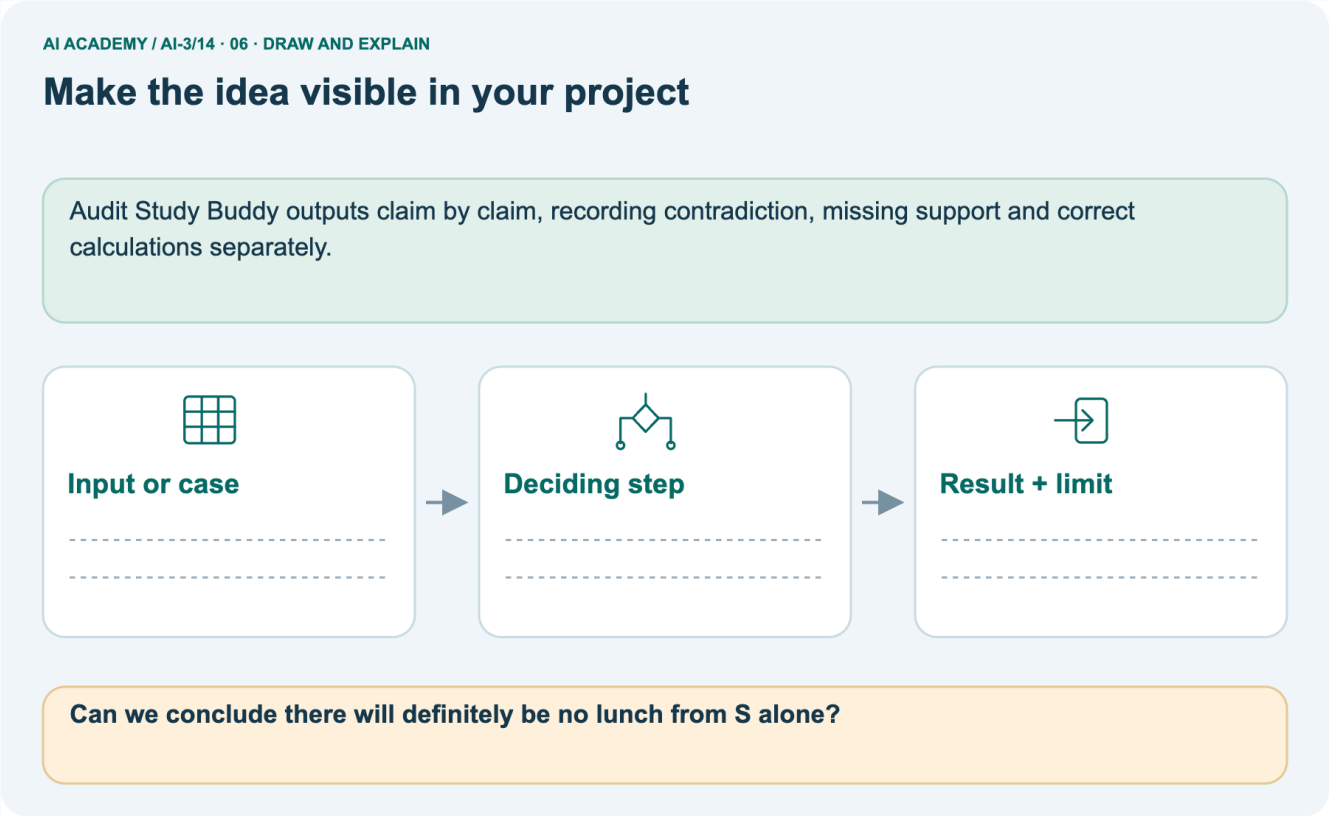
Different errors need different repairs. Recalculate the duration; keep the supported departure; remove or seek **evidence** for lunch. These categories are more informative than calling the whole answer wrong.

Arithmetic inherits assumptions such as constant speed and no stops, so avoid turning the three-hour walking calculation into a guaranteed arrival time.

Limit: The fictional plan does not establish actual speed, delays or what eventually occurred on a real trip.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Audit Study Buddy **output**s claim by claim, recording contradiction, missing support and correct calculations separately. Draw your **input**, deciding step, result and limit.

Supported question: Can we conclude there will definitely be no lunch from S alone?

Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.